

# ERYTRA EFLEXIS

## PREVENTIVE MAINTENANCE

Form version 1.0

40.000 tests  
6 Years

|                    |                |                   |     |
|--------------------|----------------|-------------------|-----|
| Instrument:        | Erytra Eflexis | Form Version:     | 1.0 |
| Serial Number:     |                | Code:             |     |
| Installation Date: |                | Maintenance Date: |     |

|                |                |
|----------------|----------------|
| Customer Site: | Customer Name: |
| Department:    |                |
| Tel:           | E-Mail:        |

|                        |         |
|------------------------|---------|
| Service Engineer Name: |         |
| Department:            |         |
| Tel:                   | E-Mail: |

|                                  |
|----------------------------------|
| Number of tests of last year PM: |
| Current number of tests:         |

### PARTS REPLACEMENT Instrument OFF

| ACTION                             | Checked/Done             | Value | Comment |
|------------------------------------|--------------------------|-------|---------|
| Probe (PR)                         | <input type="checkbox"/> |       |         |
| Toothed belt T2.5/245 (PIP Y-axis) | <input type="checkbox"/> |       |         |
| Toothed belt T2.5/245 (PIP X-axis) | <input type="checkbox"/> |       |         |
| Toothed belt T2.5/245 (CAT Y-axis) | <input type="checkbox"/> |       |         |
| Toothed belt T2.5/245 (CAT X-axis) | <input type="checkbox"/> |       |         |
| Aspiration pump VPB                | <input type="checkbox"/> |       |         |

### VISUAL / PHYSICAL INSPECTIONS & CLEANING Instrument OFF

| ACTION                                                                                                                                                                    | Checked/Done             | Value | Comment |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|---------|
| <b>GENERAL</b>                                                                                                                                                            |                          |       |         |
| General appearance                                                                                                                                                        | <input type="checkbox"/> |       |         |
| Output cables and connectors                                                                                                                                              | <input type="checkbox"/> |       |         |
| Grounding continuity test. Switch off and unplug the electrical appliance. Check with a multimeter electric continuity between earth terminal of IEC inlet appliance and: |                          |       |         |
| Screw attaching centrifuge 1 to chassis                                                                                                                                   | <input type="checkbox"/> |       |         |
| Screw attaching centrifuge 2 to chassis                                                                                                                                   | <input type="checkbox"/> |       |         |
| <b>MONITOR ARM</b>                                                                                                                                                        |                          |       |         |
| Check that the arm movement is smooth                                                                                                                                     | <input type="checkbox"/> |       |         |
| Check that the arm blocking screw prevents undesired impacts of the monitor against the instrument chassis                                                                | <input type="checkbox"/> |       |         |
| <b>DOOR</b>                                                                                                                                                               |                          |       |         |
| Check that movement is smooth                                                                                                                                             | <input type="checkbox"/> |       |         |
| Check that door does not hit during automatic opening                                                                                                                     | <input type="checkbox"/> |       |         |
| Check that the latch enters smoothly when closing                                                                                                                         | <input type="checkbox"/> |       |         |
| <b>OTHER</b>                                                                                                                                                              |                          |       |         |
| If UPS is connected, check battery level                                                                                                                                  | <input type="checkbox"/> |       |         |
| <b>CARDS MODULE GATE</b>                                                                                                                                                  |                          |       |         |
| Check that movement is smooth                                                                                                                                             | <input type="checkbox"/> |       |         |
| <b>SAMPLES AND REAGENTS ENTRY (SRE)</b>                                                                                                                                   |                          |       |         |
| Check sample tube holders                                                                                                                                                 | <input type="checkbox"/> |       |         |
| Check reagent vial/bottle lodgings                                                                                                                                        | <input type="checkbox"/> |       |         |
| <b>CARD DRAWERS</b>                                                                                                                                                       |                          |       |         |
| Check card racks centerers and retainers                                                                                                                                  | <input type="checkbox"/> |       |         |
| <b>DILUTION CUP</b>                                                                                                                                                       |                          |       |         |
| Check dilution cup drain tubing                                                                                                                                           | <input type="checkbox"/> |       |         |
| <b>CENTRIFUGE MODULE</b>                                                                                                                                                  |                          |       |         |
| Removing centrifuge pot                                                                                                                                                   |                          |       |         |

|                                                                                           |                          |
|-------------------------------------------------------------------------------------------|--------------------------|
| Clean* pot of the centrifuge                                                              | <input type="checkbox"/> |
| Check rotor and card holders status                                                       | <input type="checkbox"/> |
| Check that rotor movement is smooth                                                       | <input type="checkbox"/> |
| <b>PIP MODULE</b>                                                                         |                          |
| Check belts tension and status                                                            | <input type="checkbox"/> |
| Check chains, wiring and tubing status                                                    | <input type="checkbox"/> |
| Check/clean* barcode reader mirrors                                                       | <input type="checkbox"/> |
| <b>CAT MODULE</b>                                                                         |                          |
| Check belts tension and status                                                            | <input type="checkbox"/> |
| Check chains and wiring status                                                            | <input type="checkbox"/> |
| Check gripper pins, centerer and fingers movement/status                                  | <input type="checkbox"/> |
| Check/clean* barcode reader mirror                                                        | <input type="checkbox"/> |
| Check/clean* presence detector                                                            | <input type="checkbox"/> |
| <b>FLUIDICS MODULE</b>                                                                    |                          |
| Check status of the fluidic input/output system for each bottle pos.                      | <input type="checkbox"/> |
| Check status of the bottles                                                               | <input type="checkbox"/> |
| Perform a fluidic circuit general cleaning*                                               | <input type="checkbox"/> |
| Check tubing                                                                              | <input type="checkbox"/> |
| Check the solution lines filters status                                                   | <input type="checkbox"/> |
| Check vacuum filters status                                                               | <input type="checkbox"/> |
| Check that the main fluidic manifold has no scratches nor leakages                        | <input type="checkbox"/> |
| <b>READER MODULE</b>                                                                      |                          |
| Removing reader cover                                                                     |                          |
| Check/clean* reader pattern                                                               | <input type="checkbox"/> |
| Check/clean* reader camera optics                                                         | <input type="checkbox"/> |
| Check reader foam status                                                                  | <input type="checkbox"/> |
| <b>*If cleaning is required, please note it in the comments box.</b>                      |                          |
| <b>LUBRICATIONS</b>                                                                       |                          |
| <b>ACTION</b>                                                                             | <b>Checked/Done</b>      |
| <b>DOOR</b>                                                                               |                          |
| Lubricating general module                                                                |                          |
| Lubricate door spring and door guides (grease)                                            | <input type="checkbox"/> |
| <b>CARD DRAWERS</b>                                                                       |                          |
| Lubricating general module                                                                |                          |
| Lubricate drawers latch (grease)                                                          | <input type="checkbox"/> |
| Lubricate linear guides (grease)                                                          | <input type="checkbox"/> |
| <b>PIP MODULE</b>                                                                         |                          |
| Lubricating pipetting arm module                                                          |                          |
| Lubricate X-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate Y-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate homs linear guide (oil)                                                         | <input type="checkbox"/> |
| Lubricate probe leadscrew (grease)                                                        | <input type="checkbox"/> |
| Lubricate probe washer holder cam (grease)                                                | <input type="checkbox"/> |
| Lubricate homs rack and pinion system (grease)                                            | <input type="checkbox"/> |
| <b>CAT MODULE</b>                                                                         |                          |
| Lubricating card transport arm module                                                     |                          |
| Lubricate X-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate Y-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate gripper linear guide (oil)                                                      | <input type="checkbox"/> |
| Lubricate gripper leadscrew (grease)                                                      | <input type="checkbox"/> |
| Lubricate gripper pins (oil)                                                              | <input type="checkbox"/> |
| Lubricate gripper centering system                                                        | <input type="checkbox"/> |
| <b>FLUIDICS MODULE</b>                                                                    |                          |
| Lubricating fluidics module                                                               |                          |
| Lubricate bottle pin entry for each bottle position (grease)                              | <input type="checkbox"/> |
| <b>GENERAL</b>                                                                            |                          |
| Check the general appearance of the instrument and if necessary clean the affected parts. | <input type="checkbox"/> |
| <b>FUNCTIONAL CHECKINGS &amp; ADJUSTMENTS</b>                                             |                          |
| <b>ACTION</b>                                                                             | <b>Checked/Done</b>      |
| <b>DILUTION CUP</b>                                                                       |                          |

[Check proper stirrer operation](#)  
(PROFILE 0: 400 ± 40 rpm)

☐

#### CENTRIFUGE MODULE

[Centrifugation verification \(PROFILE 0:  
1206 rpm ± 36 rpm with duration  
of 550 s ± 10 s\)](#)

☐

#### PIP MODULE

[Check presence detector alignment](#)

☐

[Verify the CPD sensor with Leak-  
tightness test in TSS](#)

☐

[Check probe offset in TSS](#)

☐

#### CAT MODULE

[Check gripper-presence detector  
alignment](#)

☐

#### READER MODULE

[Removing reader cover](#)

[Perform reader adjustment](#)

☐

[Perform gripper adjustment in card  
reader area](#)

☐

#### SOFTWARE

[Folders inspection](#)

☐

#### COMPUTER AND TOUCHSCREEN

[SSD available space](#)

☐

[Touchscreen operability \(touch  
functionality in all screen areas\)](#)

☐

#### QUALIFICATION

| ACTION                                                                    | Checked/Done             | Value | Comment |
|---------------------------------------------------------------------------|--------------------------|-------|---------|
| <a href="#">Perform Operational Qualification (DG-<br/>AT-TSM-000150)</a> | <input type="checkbox"/> |       |         |